

1.4 Solve problems requiring the calculation of percentages

Percent means “for every one hundred”. This means a percent can be expressed as a fraction with a denominator of 100.

32% means 32 out of 100 or $\frac{32}{100}$.

Convert among decimal and percent forms

To convert a decimal to a percent:

- Multiply the decimal value by 100
- Write the percent symbol (%) after the number.

Express the following as a percent.

0.02

120

0.000552

To convert a percent to a decimal:

- Divide the percent by 100
- Remove the percent symbol

Express the following as a decimal.

106%

1.06%

0.023%

Calculate the percentage one number is of another number

- Write a fraction using the two numbers. The number after the word “of” is always the denominator and the other number is the numerator
- Use your calculator to convert the fraction to a decimal
- Express the decimal as a percent

Essentially, we are given the part and the whole. To find the percentage,

$$\text{Percentage} = \frac{\text{part}}{\text{whole}}$$

For example, 11 is what percent of 50?

$$\text{Percentage} = \frac{\text{part}}{\text{whole}}$$

$$\text{Percentage} = \frac{11}{50}$$

$$\text{Percentage} = 0.22 (\times 100)$$

$$\text{Percentage} = 22\%$$

Therefore, 11 is 22% of 50.

What percent of 75 is 12?

42 is what percent of 25?

Calculate the percent of a number

We are given the percentage and the whole. To find the part,

$$Part = percentage \times whole$$

For example, what is 20% of 35?

$$Part = percentage \times whole$$

$$Part = 20\% \times 35$$

$$Part = 0.20 \times 35$$

$$Part = 7$$

Therefore, 7 is 20% of 35.

What is 15% of \$120?

What is 29% of 106.55?

Calculate a number when a percentage of it is known

We are given the part and the percent. To find the whole,

$$Whole = \frac{part}{percentage}$$

For example, 30 is 60% of what number?

$$Whole = \frac{30}{60\%}$$

$$Whole = \frac{30}{0.60}$$

$$Whole = 50$$

Therefore, 30 is 60% of 50.

25 is 33.33% of what number?

230 is 115% of what number?

Solve. Show all steps. Clearly state your final answer.

Suppose you answered 37 out of 40 multiple choice questions correctly on a quiz. What grade did you receive on the quiz? Round your answer to the nearest percent.

Note: Assume each question is worth 1.

Suppose you buy a book for \$9.99. The HST (Harmonized Sales Tax) is 13%. What is the total cost of the book, including tax?

A recent survey showed that 35% of Canadian college campuses are in rural or remote communities. If 476 campuses are in rural or remote communities, what is the total number of Canadian college campuses?

Extra Practice

1. Convert from a percent to a decimal.

a. 3%

b. 17%

c. 200%

2. Convert from a decimal to a percent.

a. 0.009

b. 0.57

c. 1.25

3. Solve.

a. What is 50% of 120?

b. What is 120% of 80?

c. What % is 2 out of 20?

d. What % is 125 out of 50?

e. 33 is 11% of what number?

f. 2 is 8% of what number?

Solve. Show all steps. Clearly state your final answer.

a. A breakfast cereal provides 55% of the daily requirement of iron. What percentage of iron must be found in other foods?

b. A customer's purchase, before tax, comes to \$162.54. What is the total including 15% sales tax?

c. In a closing-out sale, a store reduces its prices by 70%. What is the sale price for a radio that originally cost \$40?

d. Emma's car insurance is reduced by 5% for each year that she has not had an accident. The normal cost of her insurance is \$620, but she has a 3-year "no claims bonus". How much is her insurance?

e. A new TV costs \$375. You pay a 35% deposit. How much is left to pay?